Lab 1

Introduction to Programming - Lab 1

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Answer the theory questions. Follow the instructions in the Practical section for the programming exercises.

Theory

Question 1: Summation with an accumulator

Trace `total` after each iteration and predict the final output. Explain why the initialization belongs before the loop. What changes if `total = 0` is moved inside the loop body immediately before the addition?

```
total = 0
for value in range(1, 5):
total += value
print(total)
```

Question 2: Iterating over text

Explain what value the loop variable receives on each iteration of `for ch in "aba"`. How many times does the body execute? Compare iterating directly over characters with iterating over `range(len(text))` and accessing a character by its index.

Practical

Submission: provide one complete Python 3 script for each assigned exercise, together with your test cases. Explain important decisions as code comments. Screenshots are not required.

Scope: use variables, scalar values, strings, arithmetic, conditions, and `for` or `while` loops. Built-in input/output, conversion, `range`, `len`, `abs`, `min`, `max`, and basic string operations are permitted. Do not define functions or classes, import libraries, or use lists, tuples, dictionaries or sets. Algorithmic reasoning is allowed in this Practical section.

Input and output: each numeric value is provided on its own line, in the order stated in the question. A word, command or string occupies one whole line. Print only the requested result, without input prompts. In compact examples, `/` marks a line break; the

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slash is not part of the actual input or output. Test at least one boundary case and one ordinary case in addition to the example. Multiple valid outputs are accepted only where explicitly stated.

Question 3: The longest repeated-character run

A string contains only A, C, G and T. Find the length of its longest uninterrupted run of one character. Track the current run and the best length while scanning; a later occurrence separated by another character starts a new run.

Input: Read the string.

Output: Print the longest run length.

Bounds: String length is 1 through 10000.

Example: input `ATTCGGGA`; output `3`.

Question 4: Following an odd-even process

Start with positive n. Repeatedly replace an even value by half of it and an odd value by three times it plus one. Stop as soon as the value is one. Use a while loop and include both the initial and final values in the displayed sequence.

Input: Read n.

Output: Print the visited values on one line, separated by spaces.

Bounds: 1 <= n <= 1000.

Example: input `3`; output `3 10 5 16 8 4 2 1`.

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